Protocol: Backbone Amplification of pht43e2

Background
This is a protocol to linearize pht43e2 for downstream use in Gibson assemblies.

Materials
- Q5® Hot Start High-Fidelity 2X Master Mix, *NEB*
- DNA Clean and Concentrator-5 kit, *Zymo*
- Primers
    - Forward Primer*: 5'-GGATCCTCAAGTGGTGG-3'
    - Reverse Primer: 5'-AGCACTAGTTTTTGTAATCGGC-3'
- HyClone™ HyPure Water, Molecular Biology Grade, *Cytiva*
- Template Plasmid
    - I'm using previously miniprepped samples that were grown in NEB® Stable Competent E. coli
- Thermocycler
    - Applied Biosystems 2720 Thermal Cycler

Procedure
1. Assemble the following for a 50µL PCR. Scale if necessary.
    - ☐ 19µL HyPure Water
    - ☐ 2.5µL 10 µM Forward Primer
    - ☐ 2.5µL 10 µM Reverse Primer
    - ☐ 25µL Q5® Hot Start High-Fidelity 2X Master Mix
    - ☐ 1µL of plasmid template at a concentration of ~25-50ng/µL
2. Flick PCR tube to mix and then pulse spin.
3. Transfer PCR tubes to a PCR machine and begin thermocycling.

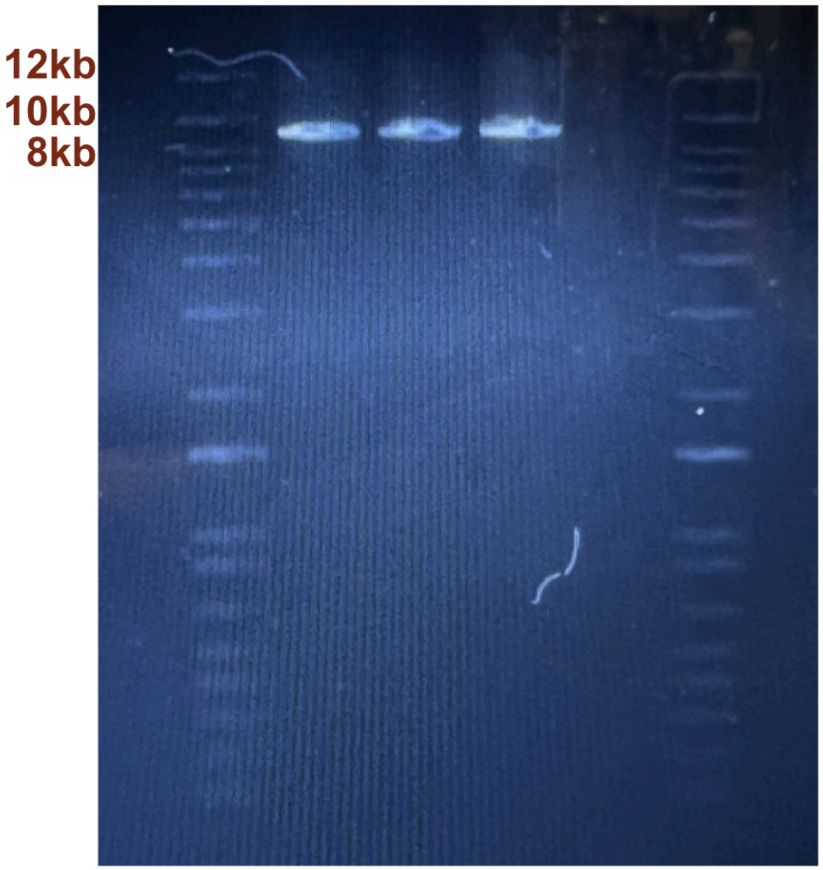
Step	***Temp***	***Time***
Initial Denaturation	98°C	2 minutes
22 Cycles	98°C	10 seconds
	63°C	30 seconds
	72°C	4 minutes 15 seconds
Final Extension	72°C	10 minutes
Hold	10°C	

4. Remove samples from PCR machine and purify using Zymo's DNA Clean and Concentrator kit. Follow kit protocol as it is described up until the elution step.

- ☐ For elution, pre-heat elution buffer to 70°C. Add 20µL of heated elution buffer to columns and incubate for 5 minutes prior to the final spin.

5. Determine yield and evaluate sample quality by nanodrop.
6. **Optional:** DpnI digest to remove any template plasmid followed by second Zymo purification. *I did not do this since I'm assuming you're using this backbone to clone single genes which will eventually be sequenced and verified.*
7. Visualize samples on gel - I run 50-60ng of sample on a 0.75% agarose gel at 75V for 85 minutes.

Example Gel Image



Three linearized pht43e2 PCR products and negative control flanked by NEB Kb+ ladders.

***A Note on the Forward Primer:** I modified the forward primer previously used for this protocol by removing 3 basepairs on the 3' end. This brings the annealing temperature closer to that of the reverse primer, reduces the %GC from 65% to 59%, and eliminates an off-target binding predicted by Snapgene.